Rebecca Kyer

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Education

Michigan State University — **East Lansing, MI**
Ph.D. Candidate; thesis advisor: Jay Strader — *Anticipated May 2027*
M.S., Astrophysics and Astronomy — *Aug 2022 – Nov 2024*

University of Washington — **Seattle, WA**
B.S., Physics and Astronomy — *Jun 2021*
Astronomy Departmental Honors

Past Research Appointments

Green Bank Observatory — **Green Bank, WV**
Graduate Student in Residence — *May 2026*

Publications

Refereed

First Author

3. "New Inclination and Orbital Period Constraints from the First Complete Census of Optical Emission Lines in Disk-State Transitional Millisecond Pulsars" **Kyer, R.**, Strader, J., Craig, P. et al. submitted to MNRAS

2. "Multiwavelength Evidence for Two New Candidate Transitional Millisecond Pulsars in the Subluminous Disk State: 4FGL J0639.1–8009 and 4FGL J1824.2+1231." **Kyer, R.**, Roy, S., Strader, J. et al. 2025, ApJ, 983, 112

1. "Monitoring the X-ray Variability of Bright X-ray Sources in M33." **Kyer, R.**, Albrecht, S., Williams, B. F. et al. 2024, ApJ, 961, 168

***n*th Author**

4. "Three New Likely Spider Millisecond Pulsar Binaries and Optical Kinematic Tracers of Intrabinary Shocks" Strader, J., **Kyer, R.**, Chomiuk, L. et al. accepted to MNRAS

3. "Investigating Four New Candidate Redback Pulsars Discovered in the Image Plane." Petrou, F., Hurley-Walker, N., McSweeney, S. et al. (incl. **Kyer, R.**) 2025, PASA, 42, 139

2. "A Survey for Radio Emission from White Dwarfs in the VLA Sky Survey." Pelisoli, I., Chomiuk, L., Strader, J. et al. (incl. **Kyer, R.**) 2024, MNRAS, 531, 1805

1. "Revisiting the Classics: On the Evolutionary Origin of the 'Fe II' and 'He/N' Spectral Classes of Novae." Aydi, E., Chomiuk, L., Strader, J. et al. (incl. **Kyer, R.**) 2024, MNRAS, 527, 9303

Non-refereed

"The Advanced X-ray Imaging Satellite Community Science Book." Koss, M., Aftab, N., Allen, S. et al (incl. **Kyer, R.**) 2025, arXiv:2511.00253

"Unlocking the Rapid X-ray Variability of Transitional Millisecond Pulsars." **Kyer, R.**, Ng, M., Coti Zelati, F. et al.

"Catching Spiders with AXIS." Coti Zelati, F., An, H., Baglio, M. C. et al. (incl. **Kyer, R.**)

Transient Name Server Classification Report No. 2024-2571, classification of SN 2024pxg as a SN II. Bostroem, K. A., **Kyer, R.**, Strader, J. et al. 2024

Classification reports in The Astronomer's Telegram (No. 16583, 16440, 16294, 17878)

Telescope Time

Principal Investigator

Green Bank Telescope GBT26A-226, 15 hr awarded (Priority B)/4.5 hr observed *2026A*

XMM-Newton AO-25 ID 098076, 35 ksec awarded (Priority C), Scientific PI *2025*

Gemini-North Fast Turnaround ID GN-2025B-FT-202, 0.38 hr *Aug 2025*

XMM-Newton AO-24 ID 096163, 57 ksec awarded (Priority C), Scientific PI *2024*

Swift Target of Opportunity ID 21032, 8.5 ksec *Aug 2024*

Swift Target of Opportunity ID 20572, 3.8 ksec *June 2024*

Co-Investigator/Observer

Green Bank Telescope 2025B Legacy ID QS402, 15 hr *2025*

200+ hours remote observing with the 4.1-m *SOAR* telescope *2023–present*

Research Advising Experience for MSU Undergraduate Students

Jadyn Elster *May 2026–present*

Analyzing long-term monitoring optical spectra of an evolved massive star with a hidden compact object companion. Jadyn is co-author of a manuscript in preparation.

Andrew Harms *May 2025–present*

Combined archival X-ray and radio data to constrain the prevalence of radio/optical nebulae associated with ultraluminous X-ray sources. Made an independent discovery of a new nebula and analyzed a follow-up optical spectrum I obtained with the Gemini Fast-Turnaround program. Andrew is preparing a manuscript for first-author publication.

Subhroja Roy (now U. Notre Dame Nuclear Astrophysics Ph.D. student) *May 2024–May 2025*

Combined archival γ-ray, X-ray and optical data to identify several strong candidate spider MSPs and analyzed follow-up optical spectroscopy. Completed astronomy senior thesis presenting two new candidate binary MSPs, one of which has been published with Subhroja as second author and the other is to be included in an upcoming publication.

Research Presentations

Rise_Time at Purdue University poster (West Lafayette, IN) *Aug 2026*

University of Tasmania Physics & Astronomy Journal Club talk (Hobart, Australia) *Aug 2026*

HEAD HMXBs and Pulsar Phenomena talk (Pasadena, CA) *June 2026*

AAS 248 Neutron Stars and Pulsar Physics dissertation talk (Pasadena, CA) *June 2026*

AAS 248 Science of Stellar Remnants press conference (Pasadena, CA) *June 2026*

Green Bank Observatory (GBO) Science Lunch talk (Green Bank, WV) *May 2026*

GBO/NRAO Radio Astronomy Bootcamp science talk (Green Bank, WV) *May 2026*
MSU Astronomy Seminar talk (East Lansing, MI) *Jan 2026*
AAS 247 AXIS Special Session invited talk (Phoenix, AZ) *Jan 2026*
AAS 247 Accretion and Evolution in X-ray Binary Systems talk (Phoenix, AZ) *Jan 2026*
Compact Objects in Michigan and Ontario (COMO) talk (Flint, MI) *Sept 2025*
Vasto Accretion Meeting talk (Vasto, Italy) *June 2025*
AXIS Community Science Meeting talk (Annapolis, MD) *May 2025*
Int'l Symposium of Science with the SOAR Telescope talk (East Lansing, MI) *May 2025*
ESAC XMM-Newton Science Workshop poster (Madrid, Spain) *June 2024*
COMO talk (Dearborn, MI) *May 2024*

Professional Service

COMO Conference Organizing Committee Chair *2026*
ESA NewAthena Science Community Working Group Member *Feb 2026–present*
AAS 247 Chambliss Poster Competition Judge *Jan 2026*
NASA AXIS Probe Compact Objects Working Group Member *Dec 2024–Mar 2026*
P-A Research Experiences for Drew Scholars Junior Director *Apr 2024–present*
Faculty Search Committee Graduate Student Representative *Feb 2024*
Stellar Mentorship Program Coordinator and Peer Mentor *Aug 2023–Aug 2026*
Astro Coffee Journal Club Organizer *Jun 2023–Aug 2026*

Outreach Activities

Letters to a Pre-Scientist Penpal *2 penpals*
Astronomy on Tap Speaker (Lansing, MI) *3 events*
Campus Observatory Public Night Volunteer (East Lansing, MI) *Aug 2022–present*

Teaching Experience

Teaching Assistant AST 208: Planets and Telescopes (30 students; MSU) *Jan–May 2024*
Teaching Assistant ISP 205L: Visions of the Universe (400 students; MSU) *Aug 2022–Dec 2023*
Instructor GEN ST 199: Astronomy First-year Interest Group (30 students; UW) *Sep–Dec 2020*

Awards and Honors

Honorable Mention, NSF Graduate Research Fellowship *Apr 2024*
Graduate Teaching Assistant Award, MSU Physics and Astronomy *Apr 2024, Apr 2023*
Mervyn Mork Excellence in Teaching Award, MSU College of Natural Science *Nov 2023*
Baer Prize for Excellence in Research, Academics and Outreach, UW Astronomy *May 2021*